LESSON 5: FLOWER, FRUITS AND SEEDS

The flower's whorls, how its parts become fruits and seeds, and their types

GENERAL BOTANY | EVALS: FLOWER, FRUITS AND SEEDS, PLANT KINGDOM

What is a flower for?

The **flower** is the plant's **reproductive part**. Its biological function is to **facilitate reproduction**, usually by providing a mechanism for the **union of sperm with eggs**. After fertilization the **ovary becomes the fruit** and the **ovules become the seeds**.

- **Smallest flower** – **Wolffia** | **Largest flower** – **Rafflesia**

THE FLOWER AND ITS FOUR WHORLS

- Four parts, outside → in: **calyx** → **corolla** → **androecium** → **gynoecium**
- **Non-essential** – **calyx, corolla** | **Essential** – **androecium, gynoecium** (the reproductive whorls)

CALYX (1ST, OUTERMOST WHORL)

- Unit: **sepal**; usually **green**; non-essential
- **Petaloid** – calyx is **brightly colored**, like petals
- **Polysepalous** – **free** sepals; **rose**
- **Gamosepalous** – **fused** sepals; **Datura**

> **Note:** The slide titled "Polysepalous" says "petals are free"; it means **sepals**.

COROLLA (2ND WHORL)

- Unit: **petal**; **variously colored**; non-essential
- **Polypetalous** – **free** petals; **rose**
- **Gamopetalous** – **fused** petals; **Datura**
- **Apetalous** – petals **absent** (rare); **Brassica rapa**

> **Note: Poly** = free (separate), **gamo** = fused, **a-** = without. **-sepalous** = calyx, **-petalous** = corolla.

ANDROECIUM AND GYNOECIUM

ANDROECIUM (3RD WHORL, MALE)

- Unit: **stamen**; essential; **male** reproductive part
- **Anther** – head of the stamen; holds the **pollen grains** (for pollination)
- **Filament** – the **stalk** that holds the anther

GYNOECIUM (INNERMOST WHORL, FEMALE)

- Unit: **carpel**; essential; **female**; after fertilization it develops into the **fruit**
- **Stigma** – top of the carpel, **sticky**; pollen sticks to it
- **Style** – the **tube** the sperm cell travels down to reach the ovary
- **Ovary** – where **fertilization** happens; produces the eggs; has the **ovary wall** and **ovules**
  - **Ovary → fruit** | **Ovary wall → outer coat of the fruit** | **Ovules → seeds**

Can-do / can't-do clues

Pollen won't stick → **stigma**. Pollen sticks but sperm never reaches the ovary → **style**. No pollen → **anther**. Anther not held up → **filament**. Fruit but no seeds → **ovules**. Seeds but no fruit coat → **ovary wall**.

TYPES OF FLOWERS

- **Complete** – **all 4 whorls**; most **bisexual** flowers are complete
- **Incomplete** – **any whorl missing**; **all unisexual flowers are incomplete**
- **Bisexual** – **androecium + gynoecium** in the same flower
- **Unisexual**: **male = staminate** (androecium only) | **female = pistillate** (gynoecium only)

FRUITS

- Fruits are found **only in angiosperms**
- **True fruit** – from the **ovary only** | **Accessory fruit** – ovary **+ receptacle, petals, sepals** (**pineapple**)
- **Simple** – from a **single ovary** (one pistil), one or more carpels; **fleshy or dry**; true or accessory
- **Aggregate** – **many pistils of the same flower**; units may be berries
- **Multiple** – pistils from **many flowers**, often with accessory parts (**mulberry**)

FLESHY FRUITS (WALL SOFT AT MATURITY)

- **Drupe** – **one seed**, from a superior ovary; **exocarp** thin skin, **mesocarp** fleshy or fibrous, **endocarp hard and stony** (cherry, peach, plum)
  - Skin = outer ovary layer, flesh = middle, pit = **inner** layer
  - **Coconut** – drupe with **fibrous** walls; liquid and solid **endosperm**
- **Berry** – ovary wall (at least its inner part) **enlarged and juicy** (grape, banana, gooseberry, tomato)
  - **Hesperidium** – **leathery rind**; **septa = number of carpels** (orange, lemon)
  - **Pepo** – **hard rind** (watermelon, gourds, squash, pumpkin, cucumber)
- **Pome** – French **pomme**, apple; **accessory**: carpels surrounded by the floral tube/receptacle; **5+ carpels**; **core from the ovary**, **pulp from the perianth** (apple, pear, mountain ash)

DRY FRUITS (COAT DRY AT MATURITY)

- **Dehiscent** – **open** along definite seams to shed seeds | **Indehiscent** – **do not open**; many are **one-seeded**

DEHISCENT

- **Legume** – **1 carpel**, splits along **both** dorsal and ventral **sutures**; two rows of ovules (pea, bean)
  - **Peanut** – one of the few legumes that **does not split**; probably because it **develops in the soil**
- **Follicle** – **1 carpel**, splits along **one suture** (larkspur, columbine, milkweed, bay magnolia)
- **Capsule** – **several carpels**; lily splits into sections = number of carpels
  - **Loculicidal** – along the **outer median line** (lily, Viola, Yucca, Datura)
  - **Poricidal** – opens by **round holes** (poppy, Papaver)
  - **Denticidal** – **small teeth** at the top (corn cockle, Agrostemma githago)
  - **Septicidal** – along the **septa**, opens at the top (Dutchman's pipe, Aristolochia)
  - **Silique** – **long, slender**, 2 carpels (mustard family, Brassicaceae/Cruciferae) | **Silicle** – **short, broad**, 2 carpels
  - **Pyxis** – **circumscissile**: the top comes off like a lid (plantain)

INDEHISCENT

- **Achene** – one seed, attached to the fruit wall at **one point**; seed free of the **pericarp** (buttercup, dandelion, sunflower)
- **Nut** – like an achene, wall **greatly thickened and hard** (beech, chestnut, oak, hazel); acorn cup = **involucre**; walnut and hickory sometimes called drupes (husk)
- **Samara** – **winged** achene; one or two seeds; **wind dispersal** (elm, maple, ash)
- **Grain (caryopsis)** – **seed coat fused to the pericarp** (wheat, corn)
- **Schizocarp** – from a compound pistil; splits into **mericarps** that look like small fruits

> **Note: Legume vs follicle** = two seams vs one. **Achene vs grain** = seed free (one point) vs fused. **Achene vs nut** = thin vs very hard wall.

SEEDS

- **Seed** – a **fertilized matured ovule** with a **seed coat**; the **propagating material**; an **embryonic plant** in a protective covering; completes reproduction in seed plants
- Three parts: **embryo**, **endosperm** (stored food), **seed coat** (testa)
- **Good seed** – **true to type**; healthy, **pure** (no inert matter, no weed seeds); **viable** (germination tested recently); **uniform**; **free from seed-borne disease**

TYPES OF SEEDS

- **By cotyledons**
  - **Monocot** – **1 cotyledon** (wheat, corn, rice); germination usually **hypogeal (except onion)**; adventitious roots; **no secondary growth**; usually **endospermic**
  - **Dicot** – **2 cotyledons** (pea, bean, mango); germination usually **epigeal (except pea and gram)**; usually **non-endospermic**; secondary growth often present; flower parts in **4s or 5s**
- **By endosperm**: **albuminous** – has endosperm (castor, poppy, custard apple) | **exalbuminous** – none (gram, pea, bean)
- **By storage**
  - **Orthodox** – survive **drying and freezing**; long storage; dried to **≤10% moisture**, stored sub-freezing (Capsicum annuum, Citrus aurantifolia, grains, legumes)

- **Recalcitrant** – **die when dried**; short storage; **chill-sensitive**; **30–70% moisture** at maturity (coffee, mango, rubber, neem, jackfruit)

- **By food**: **grains** (rice, wheat, jowar, maize, ragi) | **pulses** (grams, peas, beans) | **oilseeds** (groundnut, sesame, mustard, coconut, sunflower)

Note: Hypogeal = cotyledons stay **below** ground (most monocots); **epigeal** = cotyledons come **above** (most dicots).

SUMMARY TABLE: PARTS OF THE FLOWER

Whorl	Position	Unit	Essential?	Notes
Calyx	1st, **outermost**	**sepal**	non-essential	usually green; **petaloid** if brightly colored; **poly-** (free, rose) or **gamosepalous** (fused, Datura)
Corolla	2nd	**petal**	non-essential	variously colored; **polypetalous** (rose), **gamopetalous** (Datura), **apetalous** (Brassica rapa)
Androecium	3rd	**stamen**	**essential**, male	**anther** (pollen) + **filament** (stalk)
Gynoecium	**innermost**	**carpel**	**essential**, female	**stigma** (sticky), **style** (tube), **ovary** → **fruit**; ovary wall → fruit coat; ovules → **seeds**

SUMMARY TABLE: FLESHY FRUITS

Type	Key feature	Examples
Drupe	**one seed** in a **stony endocarp**; fleshy or fibrous mesocarp; thin exocarp	cherry, peach, plum, **coconut** (fibrous)
Berry	ovary wall becomes **juicy**	grape, banana, gooseberry, tomato
Hesperidium	berry with a **leathery rind**; **septa = carpels**	orange, lemon
Pepo	berry with a **hard rind**	watermelon, squash, pumpkin, cucumber
Pome	**accessory**; core from **ovary**, pulp from **perianth**; 5+ carpels	apple, pear, mountain ash

SUMMARY TABLE: DRY FRUITS

Type	Opens?	Key feature	Examples
Legume	dehiscent	1 carpel; splits along **both sutures**	pea, bean (**peanut** stays shut)
Follicle	dehiscent	1 carpel; splits along **one suture**	larkspur, columbine, milkweed
Capsule	dehiscent	**several carpels**: loculicidal (median line), **poricidal** (holes), **denticidal** (teeth), **septicidal** (septa), **silique** (long), **silicle** (short), **pyxis** (lid)	lily; poppy; corn cockle; Dutchman's pipe; mustard; plantain
Achene	indehiscent	one seed attached at **one point**	buttercup, dandelion, sunflower
Nut	indehiscent	like an achene, **very hard wall**	oak (acorn), chestnut, hazel
Samara	indehiscent	**winged**; wind dispersal	elm, maple, ash
Grain (caryopsis)	indehiscent	**seed coat fused to pericarp**	wheat, corn
Schizocarp	splits into pieces	breaks into **mericarps**	compound pistil fruits

SUMMARY TABLE: SEED TYPES

Type	Key feature	Examples
Monocot	1 cotyledon; **hypogeal** (not onion); endospermic	wheat, corn, rice
Dicot	2 cotyledons; **epigeal** (not pea, gram); non-endospermic	pea, bean, mango
Albuminous	**has endosperm**	castor, poppy, custard apple
Exalbuminous	**no endosperm**	gram, pea, bean
Orthodox	survives drying/freezing; **≤10% moisture**; long storage	Capsicum annuum, Citrus aurantifolia, grains, legumes
Recalcitrant	**dies when dried**; 30–70% moisture; chill-sensitive	coffee, mango, rubber, neem, jackfruit

LESSON 6: PLANT KINGDOM

How plants are classified, from algae and mosses to ferns, cone-bearers and flowering plants

GENERAL BOTANY | EVALS: FLOWER, FRUITS AND SEEDS, PLANT KINGDOM

What makes something a plant?

All plants are **producers** (photosynthesis), have **eukaryotic cells with cell walls** (cellulose) and have a **cuticle**. Kingdom Plantae = **eukaryotic, multicellular, autotrophic**, with **chloroplasts** holding **chlorophyll**. Sizes run from **duckweed (about 10 mm)** to the **giant redwood (about 100 m)**.

CLASSIFYING PLANTS

- Three criteria: **plant body** (root, stem, leaves?) | **vascular system** (xylem, phloem?) | **seed formation** (flowers and seeds; **naked** or **in a fruit**?)
- **Nonvascular** (mosses, liverworts) | **Vascular and seedless** (ferns, club mosses, horsetails) | **Vascular with seeds** (gymnosperms, angiosperms)
- **Cryptogams** – **non-flowering, non-seed-bearing**: Thallophyta, Bryophyta, Pteridophyta
- **Phanerogams** – **flowering, seed-bearing**: Gymnosperms, Angiosperms
- **Seed** – an **embryo plus a food supply** inside a protective covering

THALLOPHYTA

- Body is a **thallus**: **no true roots, stems or leaves**; **no vascular tissue**
- Mostly **aquatic** or in moist places; take in water and nutrients **directly** from the surroundings
- **Autotrophic** (the algae); some members (**fungi**) are **heterotrophic**
- **Starch** reserve; **cellulose** wall; **simple, single-celled sex organs**; **no embryo** after fertilization
- **Volvox** (colonial) | **Spirogyra**, **Ulothrix** (filamentous) | **Cladophora** (branched) | **Chara** | **Ulva** (sea lettuce, eaten) | **Fucus** (brown) | **Polysiphonia** (red; red algae give **agar**)

Note: The slide tree also lists bacteria and fungi under Thallophyta; the notes describe thallophytes as mostly **algae**.

BRYOPHYTA

- **No vascular tissue**; **root-like (rhizoids), stem-like and leaf-like** parts; the **"amphibians of the plant kingdom"**: land plants that need **water** to reproduce
- Moist, shady places; **mosses, hornworts, liverworts** (Marchantia, Funaria, Sphagnum, Anthoceros)
- Reproduce by **spores**; **sperm swim** through a water film; can **dry out and revive**
- **Moss cycle**: capsule makes **spores (meiosis)** → spore grows a **protonema** → leafy **gametophyte** (dominant) makes sperm and eggs → water → **zygote** → **sporophyte** (stalk + capsule), which stays on the **female gametophyte** and depends on it
- **Liverwort** – sporophyte looks like a **tiny palm tree**; gametophyte **leafy and flattened**; **rhizoids** anchor

PTERIDOPHYTA

- **Seedless vascular** plants (ferns, horsetails, club mosses) with true **roots, stems and leaves**: Selaginella, Equisetum, Pteris
- Reproduce by **spores** in **sporangia**, often under the **fronds**
- **Fern parts**: **sporophyte** (the leafy fern) | **frond** (leaf, photosynthesis) | **rhizome** (stem; roots and fronds grow from it) | **fiddlehead** (young coiled frond)
- **Fern cycle**: sporophyte (**diploid**) makes spores by **meiosis** → small, heart-shaped **gametophyte (prothallus, haploid)** with **antheridia (sperm)** and **archegonia (egg)** → sperm with **flagella** swim through water → **zygote** → young **sporophyte**
- **Alternation of generations** – sporophyte and gametophyte take turns; **ferns: sporophyte dominates**; **mosses: gametophyte dominates**
- **Homosporous** – one spore type, bisexual gametophyte (**Psilotum**) | **Heterosporous** – **microspores (male)** and **megaspores (female)** (**Selaginella**), a step toward seeds

Note: Correction from the notes: the sperm swims toward an **unfertilized** egg; the egg is fertilized after the sperm fuses with it.

SEED PLANTS

GYMNOSPERMS

- Well-differentiated body, vascular tissue and **naked seeds** (not in a fruit), housed in **cones**
- **Conifers, cycads, ginkgoes** (Cycas, Pinus, Ephedra); the **bristlecone pine** is the **oldest living organism**

ANGIOSPERMS

- **Flowering** vascular plants with seeds **enclosed in a fruit**; the **majority of all plants**; trees, herbs, shrubs, bushes; the **most advanced and beneficial** group (mango, rose, tomato, onion, wheat, maize)
- Sizes: **Wolffia** about 0.1 cm | **Eucalyptus** about 100 m
- Divided by cotyledons into **monocots** and **dicots**

SUMMARY TABLE: PLANT GROUPS AT A GLANCE

Group	Body	Vascular?	Reproduces by	Examples
Thallophyta	**thallus**: no true roots, stems or leaves	no	simple single-celled sex organs; **no embryo**	Volvox, Spirogyra, Ulva, Fucus
Bryophyta	**root-like, stem-like, leaf-like** parts; **rhizoids**	no	**spores**; needs **water**	Marchantia, Funaria, Sphagnum
Pteridophyta	true **roots, stems, leaves**	**yes**	**spores** in sporangia; needs **water**	Selaginella, Equisetum, Pteris
Gymnosperms	well differentiated	**yes**	**naked seeds** in **cones**	Cycas, Pinus, Ephedra
Angiosperms	well differentiated	**yes**	seeds **in a fruit** (flowers)	mango, rose, tomato, wheat, maize

SUMMARY TABLE: MOSS AND FERN

Feature	Moss (bryophyte)	Fern (pteridophyte)
Dominant stage	**gametophyte** (green, leafy)	**sporophyte** (the leafy fern)
Vascular tissue	none	xylem and phloem
Anchored by	**rhizoids**	true roots (and a **rhizome**)
From the spore	**protonema**, then the leafy gametophyte	**prothallus** (small, heart-shaped gametophyte)
Sporophyte	stalk and **capsule**, stays on the female gametophyte	independent plant with **fronds**; **sporangia** under the fronds
Fertilization	sperm swim through a **water film**	flagellated sperm swim through a **water film**

SUMMARY TABLE: MONOCOT AND DICOT

Feature	Monocot	Dicot
Cotyledons	**one**	**two**
Leaves	**parallel** veins	**reticulate** (netted) veins
Roots	**adventitious**	**tap root**
Floral parts	in **threes**	in **fours or fives**
Vascular bundles	**scattered** in the ground tissue	in a **ring**
Pollen grain	**one** pore or furrow	**three** pores or furrows
Examples	banana, sugarcane, lily, wheat, maize	grape, sunflower, tomato, mango, rose

END OF LESSON